wavefront in the plane of the paper will be a circle. Once again, both the ordinary and the extraordinary rays travel along the same direction. It may be mentioned that Figs. 22.24 (a) and (b) correspond to the same configuration; in both cases the optic axis is parallel to the surface. The figures represent two different cross-sections of the same set of spherical and ellipsoidal wavefronts.

Now, corresponding to Figs. 22.24 (a) and (b), if the incident wave is polarized perpendicular to the optic axis, it will propagate as an *o*-wave with velocity c/n_o. On the other hand, if the incident wave is polarized parallel to the optic axis, it will propagate as an *e*-wave with velocity c/n_e. In Fig. 22.24 (c) the optic axis is normal to the surface and both waves will travel with the same velocity.

Notice that in the configuration shown in Figs. 22.24 (a) and (b), although both the waves travel in the same direction, they propagate with different velocities. This phenomenon is used in the fabrication of quarter and half wave plates (see Sec. 22.6). On the other hand, in the configuration shown in Fig. 22.24(c), both the waves not only travel in the same direction but they also propagate with the same velocity.

22.5.2 Oblique Incidence

We next consider the case of a plane wave incident obliquely on a negative uniaxial crystal [see Fig. 22.25(a)]. Once again we use Huygens' principle to determine the shape of the refracted wavefronts. Let BD represent the incident wavefront. If the time taken for the disturbance to reach the point F from D is t, then with B as center we draw a sphere of radius $(c/n_o)t$ and an ellipsoid of revolution of semi-minor and semi-major axes $(c/n_o)t$ and $(c/n_e)t$ respectively; the semi-minor axis is along the optic axis. From the point F we draw tangent planes FO and FE to the sphere and the ellipsoid of revolution respectively. These planes would represent the refracted wavefronts corresponding to the ordinary and the extraordinary rays respectively. If the points of contact are O and E, then the ordinary and extraordinary refracted rays will propagate along BO and BE respectively; this can also be shown using Fermat's principle (see Sec. 2.5). The directions of vibration of these rays are shown by dots and small lines respectively and are obtained by using the general rules discussed earlier. The shape of the refracted wavefronts corresponding to the particular case of $\alpha = 0$ and $\alpha = \pi/2$ can be obtained very easily. Figure 22.24(b) corresponds to the case when the optic axis is normal to the plane of incidence. The sections of both the wavefronts will be circles; consequently, the extraordinary ray will also satisfy Snell's law and we will have

$$\frac{\sin i}{\sin r} = n_e \quad \text{(for the } e\text{-ray when the optic-axis is normal to the plane of incidence)} \qquad \textbf{(22.46)}$$

Of course, for the ordinary ray we will *always* have

$$\frac{\sin i}{\sin r} = n_o \qquad \textbf{(22.47)}$$

22.6 INTERFERENCE OF POLARIZED LIGHT: QUARTER WAVE PLATES AND HALF WAVE PLATES

LO 4

In the previous section we had considered how a plane wave (incident on a doubly refracting crystal) splits up into two

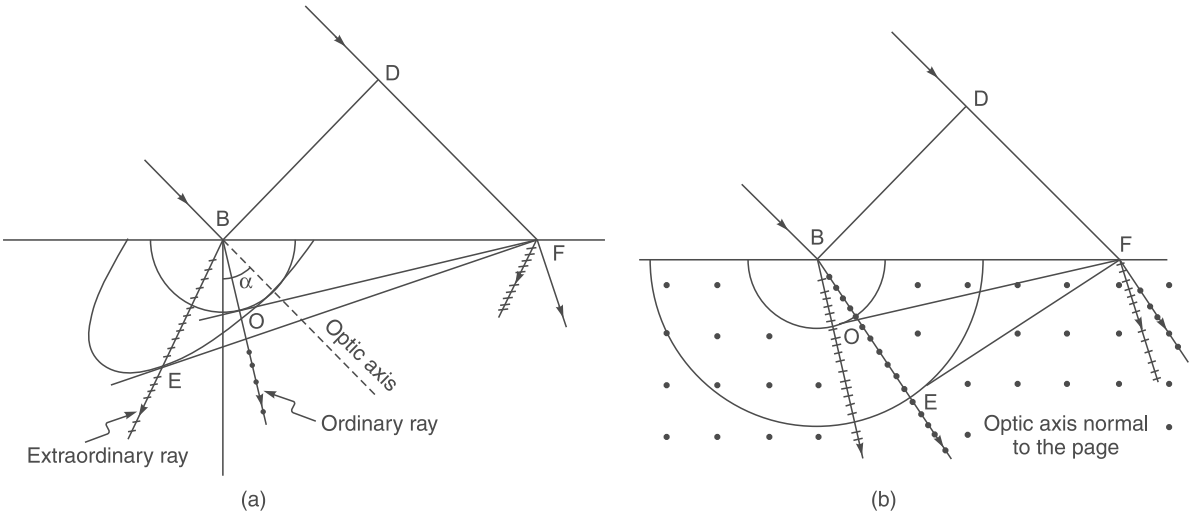


Fig. 22.25 Refraction of a plane wave incident obliquely on a negative uniaxial crystal. In (a), the direction of the optic axis is along the dashed line. In (b), the optic axis is perpendicular to the plane of the paper.

waves each characterized by a certain state of polarization. The direction of vibration associated with the ordinary and extraordinary waves is obtained by using the recipe given by Eqs. (22.42) and (22.43). In this section, we will consider the normal incidence of a plane-polarized beam on a calcite crystal whose optic axis is parallel to the surface of the crystal as shown in Fig. 22.26. We will study the state of polarization of the beam emerging from the crystal. We will assume the *y*-axis to be along the optic axis. Now, as discussed in the previous section, if the incident beam is *x*-polarized the beam will propagate as an ordinary wave and the extraordinary wave will be absent. Similarly, if the incident beam is *y*-polarized the beam will propagate as an extraordinary wave and the ordinary wave will be absent– these are the modes of the crystal. For any other state of polarization of the incident beam, both the extraordinary and the ordinary components will be present. For a negative crystal like calcite $n_e < n_o$ and the *e*-wave will travel faster than the *o*-wave; this is shown by putting *s* (slow) and *f* (fast) inside the parenthesis in Fig. 22.26. For a positive crystal like quartz $n_e > n_o$ and the *e*-wave will travel faster than the *o*-wave.

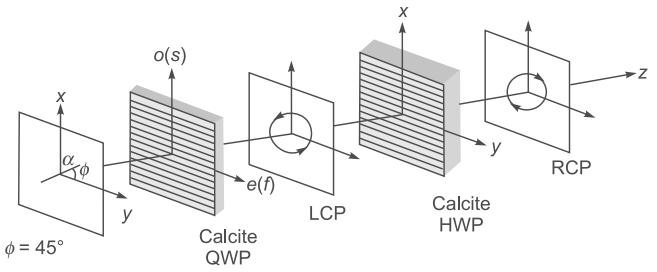


Fig. 22.26 A linearly polarized beam making an angle 45° with the *y*-axis gets converted to a LCP after propagating through a calcite QWP; further, an LCP gets converted to a RCP after propagating through a calcite HWP. The optic axis in the QWP and HWP is along the *y*-direction as shown by lines parallel to the *y*-axis.

Let the electric vector (of amplitude E_0) associated with the incident polarized beam make an angle ϕ with the *y*-axis; in Fig. 22.26, ϕ has been shown to be equal to 45°—but for the time being we will keep our analysis general and assume ϕ to be an arbitrary angle. Such a beam can be assumed to be a superposition of two linearly polarized beams (vibrating in phase), polarized along the *x*- and *y*-directions with amplitudes $E_0 \sin\phi$ and $E_0 \cos\phi$ respectively. The *x*-component (whose amplitude is $E_0 \sin\phi$) passes through as an ordinary beam propagating with velocity $E_0 \cos\phi$. The *y*-component (whose amplitude is $E_0 \sin\phi$) passes through as an extraordinary beam propagating with velocity c/n_o. Since $n_e \neq n_o$ the two beams will propagate with different velocities and, as such, when they come out of the crystal, they will not be in phase. Consequently, the emergent beam (which will be a superposition of these two beams) will be, in general, elliptically polarized.

Let the plane $z = 0$ represent the surface of the crystal on which the beam is incident. The *x*- and *y*-components of the incident beam can be written in the form

$$\begin{aligned} E_x &= E_0 \sin\phi \cos(kz - \omega t) \\ E_y &= E_0 \cos\phi \cos(kz - \omega t) \end{aligned} \qquad \textbf{(22.48)}$$

where $k\,(=\omega/c)$ represents the free-space wave number. Thus, at $z = 0$, we will have

$$E_x(z=0) = E_0 \sin\phi \cos\omega t; \quad E_y(z=0) = E_0 \cos\phi \cos\omega t$$

Inside the crystal, the *x*-component will propagate as an ordinary wave (with velocity c/n_o) and the *y*-component will propagate as an extraordinary wave (with velocity c/n_e)

$$E_x = E_0 \sin\phi \cos(n_o kz - \omega t) \qquad \text{(ordinary wave)}$$

$$E_y = E_0 \cos\phi \cos(n_e kz - \omega t) \qquad \text{(extraordinary wave)}$$

If the thickness of the crystal is *d*, then at the emerging surface, we will have

$$\begin{aligned} E_x &= E_0 \sin\phi \cos(\omega t - \theta_o) \\ E_y &= E_0 \cos\phi \cos(\omega t - \theta_e) \end{aligned}$$

where $\theta_o = n_o k d$ and $\theta_e = n_e k d$. By appropriately choosing the instant $t = 0$, the components may be rewritten as

$$\begin{aligned} E_x &= E_0 \sin\phi \cos(\omega t - \theta) \\ E_y &= E_0 \cos\phi \cos\omega t \end{aligned} \qquad \textbf{(22.49)}$$

where $\theta = \theta_o - \theta_e = k\,d\,(n_o - n_e) = \dfrac{\omega}{c}(n_o - n_e)\,d$ **(22.50)**

represents the phase difference between the ordinary and the extraordinary beams. Clearly, if the thickness of the crystal is such that $\theta = 2\pi, 4\pi, 6\pi, \ldots$, the emergent beam will have the same state of polarization as the incident beam. Now, if the thickness *d* of the crystal is such that $\theta = \pi/2$, the crystal is said to be a quarter wave plate (usually abbreviated as QWP)—a phase difference of $\pi/2$ implies a path difference of a quarter of a wavelength. On the other hand, if the thickness of the crystal is such that $\theta = \pi$, the crystal is said to be a half wave plate (usually abbreviated as HWP).

Example 22.6 As an example, let us consider the case when $\phi = \pi/4$ and $\theta = \pi/2$, i.e., the *x*- and *y*-components of the incident wave have equal amplitudes and the crystal introduces a

phase difference of $\pi/2$ (see Fig. 22.24). Thus, for the emergent beam we have

$$E_x = \frac{E_0}{\sqrt{2}} \sin\omega t \; ; E_y = \frac{E_0}{\sqrt{2}} \cos\omega t \qquad (22.51)$$

If we use a method similar to that described in Example 22.1, we will find that a wave described by the above equation represents a left circularly polarized wave. In order to introduce a phase difference of $\pi/2$, the thickness of the crystal should have a value given by the following equation:

$$d = \frac{c}{\omega(n_o - n_e)} \frac{\pi}{2} = \frac{1}{4} \frac{\lambda_0}{(n_o - n_e)} \qquad (22.52)$$

where λ_0 is the free-space wavelength. For calcite, for $\lambda_0 = 5893$ Å and at 18 C, $n_o = 1.65836$ and $n_e = 1.48641$. Substituting these values, we obtain

$$d = \frac{5.893\times10^{-7}}{4\times0.17195} \text{ m} \approx 0.000857 \text{ mm}$$

Thus a calcite QWP (at $\lambda_0 = 5893$ Å $= 0.5893$ μm) will have a thickness of 0.000857 mm and will have its optic axis parallel to the surface; such a QWP will introduce a phase difference of $\pi/2$ between the ordinary and the extraordinary components at $\lambda_0 = 5893$ Å. It should be pointed out that if the thickness is an odd multiple of the above quantity, i.e., if

$$d = (2m+1) \frac{\lambda_0}{4(n_o - n_e)} \; ; m = 0, 1, 2, \ldots \qquad (22.53)$$

then in the example considered above (i.e., when $\phi = \pi/4$), it can easily be shown that the emergent beam will be left circularly polarized for m = 0, 2, 4, … and right circularly polarized for m = 1, 3, 4, …

We next consider the case when the linearly polarized beam (with $\phi = \pi/4$) is incident on a HWP so that $\theta = \pi$, i.e., the x- and y-components of the incident wave have equal amplitudes and the crystal introduces a phase difference of π (see Fig. 22.27). Thus, for the emergent beam we have

$$E_x = -\frac{E_0}{\sqrt{2}} \cos\omega t \quad ; \quad E_y = \frac{E_0}{\sqrt{2}} \cos\omega t$$

which represents a linearly polarized wave with the direction of polarization making an angle of 135° with the y-axis (see Fig. 22.27). If we now pass this beam through a calcite QWP, the emergent beam will be right circularly polarized as shown in Fig. 22.27. On the other hand, if a left circularly polarized is incident normally on a calcite HWP, the emergent beam will be right circularly polarized as shown in Fig. 22.26.

Thus, for a HWP the thickness (for a negative crystal) would be given by

$$d = (2m+1) \frac{\lambda_0}{2(n_o - n_e)}$$

We may mention that if the crystal thickness is such that if $\theta \neq \pi/2, \pi, 3\pi/2, 2\pi, \ldots$ the emergent beam will be elliptically polarized. For a positive crystal (like quartz), $n_e > n_o$ and Eq. (22.49) should be written in the form

$$\begin{aligned} E_y &= E_0 \sin\phi \cos(\omega t + \theta') \\ E_z &= E_0 \cos\phi \cos\omega t \end{aligned} \qquad (22.54)$$

where

$$\theta' = \frac{\omega}{c} d(n_e - n_o)$$

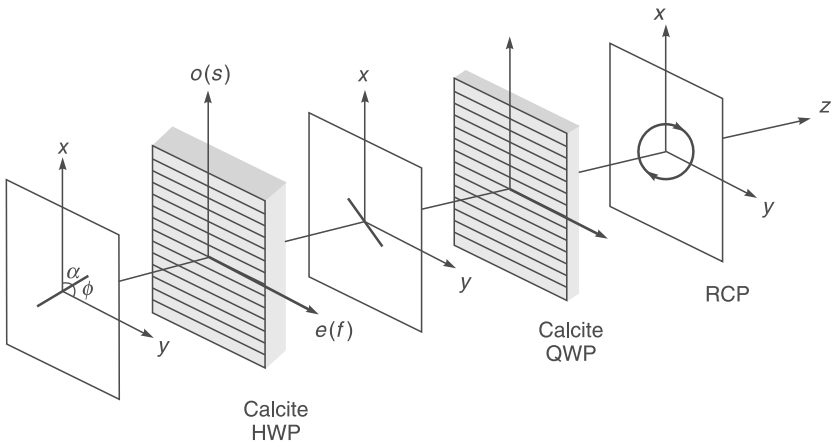


Fig. 22.27 If the linearly polarized beam making an angle 45° with the y-axis is incident on a HWP, the plane of polarization gets rotated by 90°; this beam gets converted to a RCP after propagating through a calcite QWP. The optic axis in the QWP and HWP is along the y-direction as shown by lines parallel to the z-axis.